

A Multi-Modal Machine Learning Gatekeeper for Adaptive Eavesdropper Detection in BB84 Quantum Key Distribution



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BSc (Hons) Computer Science

A dissertation submitted for the degree of
Master of Science in Data Science and Artificial Intelligence

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September, 2026

Declaration

I declare that the work presented in this dissertation is, to the best of my knowledge and belief, original and my own work. The material has not been submitted, either in whole or in part, for a degree at this, or any other university. Estimated word count is: **9835**

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Abstract

This work investigates whether a multi-modal, temporally-aware machine learning classifier can improve eavesdropper detection in BB84 quantum key distribution (QKD) networks. We focus specifically on the performance of a classifier when an adaptive eavesdropper heavily suppresses their Quantum Bit Error Rate (QBER) signature, which is the most commonly monitored signature by existing single-observable detectors, whilst leaving trace information in other channel statistics. We showed that using a Long Short-Term Memory (LSTM) network outperforms non-sequential classical baselines in aggregate performance on a fixed dataset and its advantage is most apparent when an attacker suppresses their signatures on the quantum channel. We created a synthetic dataset, modelling QBER, SPAD timing jitter and dark-count rate under an intercept-resend attack with sparsely distributed episodes and variable intensities, including a stealthy attack subtype employed for 40% of all attacks. This stealthy attack highly suppresses QBER and moderately suppresses jitter, whilst leaving dark-count rate at maximum intensity. The LSTM was benchmarked against Random Forest and Logistic Regression baselines across five random seeds on a fixed dataset. To test if these findings hold beyond a single dataset, the models were additionally trained on a pool of ten independently generated datasets and evaluated separately on eight unseen datasets. The LSTM achieved a statistically significant advantage in aggregate ROC-AUC score over both baseline models on a fixed dataset, and significantly outperformed both baselines in detecting stealthy attacks, with the LSTM's recall being 90.0%, substantially beating Random Forest at 62.6% and Logistic Regression at 82.6%. However, when the models were evaluated on unseen datasets, the LSTM's aggregate advantage over Random Forest was no longer statistically significant, whilst its advantage in recall from stealthy attacks remained robust against both baseline models. These results show that temporal awareness provides a measurable and robust advantage, specifically against an adaptive adversary who suppresses their signature, even where aggregate performance show very small differences between models. Our research has direct implications for how QKD anomaly detection systems should be evaluated, as aggregate metrics alone can conceal the kind of vulnerability a real adaptive eavesdropper will seek to exploit.

Acknowledgements

I would like to thank my supervisor, Dr. Richard Jiang, for his guidance and support throughout this project.

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Chapter 1

Introduction

Modern security for digital communication increasingly relies on public-key encryption such as RSA or elliptic curve cryptography, where the security is dependent on the computational difficulty of problems like integer factorisation. Quantum computers threaten to invalidate these systems, through quantum algorithms such as Shor’s algorithm which can solve these problems efficiently, leaving currently deployed public-key infrastructure insecure against a sufficiently powerful quantum approach. Quantum Key Distribution (QKD) offers a solution to this problem, by deriving its security directly from the laws of quantum mechanics, rather than computational hardness assumptions that may eventually be surpassed. Protocols such as BB84 offer information-theoretic security that holds regardless of an adversary’s computational power.

QKD’s security depends on the ability to detect the presence of an eavesdropper on the quantum channel. The most common technique is to monitor the rate of mismatched bits between the sender and receiver, which is called the Quantum Bit Error Rate (QBER). If the QBER rises above a predetermined threshold, then the key generation is aborted on the assumption that the rise in QBER is due to an eavesdropper being present and not just ordinary hardware noise. This strategy assumes that an eavesdropper’s presence will manifest primarily as a rise in QBER. However, an adaptive adversary could plausibly suppress their QBER signature and evade detection by staying below the agreed-upon threshold, for example, by intercepting only a fraction of transmitted pulses rather than the entire stream, introducing fewer errors, whilst still recovering partial information about the key[3]. This strategy would still disturb the channel in ways that leave measurable trace information in other channel statistics, such as timing jitter and dark-count rate. Existing workExisting work on temporal or multi-observable QKD anomaly detection has not modelled an eavesdropper who suppresses their QBER signature while leaving trace information in other channel statistics, within a intercept-resend attack [32]. This research makes two contributions to address this gap: (i) we show that a multi-modal, temporally aware classifier outperforms non-sequential models, in aggregate, on a synthetically generated telemetry

data of an intercept-resend quantum channel (**H1**), (ii) the advantage of these multi-modal temporally aware classifiers is seen most when a stealthy attack strategy is implemented that suppresses the QBER channel specifically (**H2**), where existing single-observable detectors are least equipped to handle.

This project aims to determine whether machine learning models that monitor how multiple physical channels change over time can be efficient QKD gatekeepers and improve on the standard QBER threshold detection. Specifically, we investigate two research questions: (1) does a temporally-aware classifier outperform non-sequential classical baseline models at detecting an intercept-resend eavesdropper in aggregate, across a range of attack intensities and strategies? and (2) is there a substantial advantage specifically against an adaptive eavesdropper who deliberately suppresses her QBER signature, where existing single-observable detectors are least equipped to detect? As QKD networks move from simulation and small network deployment into wider commercial deployment, the anomaly detection systems that safeguard them will need to be evaluated on their performance, not just against naive attackers, but on their robustness to attackers who adapt their strategy to circumvent security strategies specifically implemented against them.

The structure of this report is organised as follows: Chapter 2 reviews the relevant background literature on the BB84 protocol, QBER-based eavesdropper detection, and prior machine-learning approaches to QKD security. Chapter 3 describes the synthetic telemetry generation process, the data preprocessing and sequence formatting pipeline, and the LSTM and classical baseline model architectures. Chapter 4 presents experimental setup, training hyper-parameters, and the evaluation metrics used. Chapter 5 presents the aggregate and stratified performance results for all three models, as well as a cross-dataset generalisation test evaluating the models on unseen data. Chapter 6 interprets these results according to H1 and H2, discusses the validity of the approach and outlines the project’s scope and limitations and finally Chapter 7 concludes our findings and contributions as well as proposes avenues for future work.

Chapter 2

Literature Review

Classic security methods rely on complex mathematics to encrypt data. These algorithms make data resistant to most unauthorised access by modern computers. However, with the steadfast development of quantum computing, these highly complex mathematical problems that were once beyond real-time computational feasibility are now being solved efficiently. Standard public key cryptographic methods, such as RSA integer factorisation[23] and Elliptic curve discrete logarithms[13, 20], depend on the assumed intractability of one-way trapdoor functions. Despite this, publicly available and widely used methods like Shor’s algorithm [27] are making once standard protocols obsolete, leading to the immediate importance of innovative quantum-resistant solutions.

Grover’s algorithm [9] shows a related threat: it quadratically speeds up brute-force guessing of symmetric encryption functions, such as Advanced Encryption Standard (AES). This is less of a threat compared to Shor’s algorithm as it can usually be mitigated through doubling the length of key used, as opposed to requiring a different cryptographic approach, but it highlights the need for improved security as quantum approaches mature.

Recent research focuses on two main solutions: Post-Quantum Cryptography (PQC) and Quantum Key Distribution (QKD). PQC involves using the underlying software of classic computers to create algorithms for securing data and communication that are safe from attacks even from future quantum computers. On the other hand, QKD bypasses computational boundaries through the utilisation of quantum mechanics to create information-theoretic security (ITS) [31, 28].

BB84 [2] is a prepare-and-measure protocol where a sender (Alice) encodes each bit of a candidate key into the state of a single photon, chosen randomly from one of two bases, either the conjugate rectilinear ($\{|0\rangle, |1\rangle\}$) or the diagonal ($\{|+\rangle, |-\rangle\}$) bases. This gets transmitted to a receiver (Bob) who then measures each photon after randomly choosing a basis. This random choice is where the protocol gets its security as quantum mechanics guarantees that measuring a qubit with the incorrect basis outputs a uniformly random result and makes it impossible to return the qubit to its original state [21]. This is known as the

no-cloning theorem [33]. Theoretically, this means that an eavesdropper cannot passively observe the channel as any attempt to take any measurements from the channel using the wrong basis will result in an error that Bob can detect.

After this transmission, Bob and Alice compare the basis used for each photon over a secure classical channel, discarding all instances where their bases did not match. This is known as sifting, and keeps measurements that correctly match which becomes the raw key. Following this, a random sample of the sifted key is publicly compared between Alice and Bob to estimate the Quantum Bit Error Rate (QBER), then the sample is discarded, as it has been exchanged over a public classical channel, with the remaining unrevealed bits of the sifted key forming the final key [34]. QBER is the proportion of sifted bits that do not match between Alice and Bob. In ideal conditions the QBER of a channel should only represent hardware imperfections, such as detector noise, decoherence and channel loss, as well as staying at a low measurable baseline value. In an intercept-resend attack, such as the one investigated in this research, the attacker (Eve) must measure each intercepted photon and randomly choose a basis to send a new photon to Bob; this new photon will be detected by Bob with probability of 0.5, which produces an additional theoretical maximum QBER of 25% [2, 19].

This means the QBER is a measurable security signal inherently in BB84 implementations, as if the QBER rises above a certain threshold that is previously agreed by sender and receiver [14, 17], the protocol can abort and discard the key, assuming that the noise signatures on the channel are no longer hardware based and that there is an eavesdropper. This is the limitation of the BB84 protocol that this research investigates. Monitoring only QBER assumes that an eavesdropper will not attempt an adaptive attack strategy such as trying to suppress their channel signature. Recent literature increasingly disputes this assumption. Brassard et al. [3] show that an eavesdropper that only intercepts a fraction of transmitted qubits induces less QBER disturbance, while Saxena et al. [24] show that an eavesdropper exploiting a compromised random number generator can recover a complete key with minimal rise in QBER, evading detection entirely. This research tests this assumption directly by evaluating models against an eavesdropper who deliberately suppresses their channel signatures.

Networks used for commercial and professional deployment are much less perfect than basic simulations and models. Recent studies on advanced networks track broader metrics to ascertain all information that should be considered when investigating possible attacks [28, 15]. Dasari et al. [6] investigated channel efficiency tolerances and multi-protocol phase visibility degradation during photon-number splitting attacks. However, this investigation treated phase visibility and raw QBER as static variables which do not reflect how a real attacker will use dynamic strategies to match normal environmental fibre loss and go undetected.

Jha et al. [11] investigated the use of deep neural networks for QKD; however using networks requires large datasets to learn channel noise profiles, which are impractical for

real-time use. Jha et al. implemented a Gaussian process regression (GPR) to map distance, background noise and transmittance to predict the Quantum Bit Error rate (QBER). This resulted in a prediction error less than 10^{-3} using fewer than 250 data samples whilst maintaining uncertainty confidence intervals of ± 2 (standard deviations). Although this implementation solves the problem of small sample training, it fails to generalise to non-static channel environments, where active and dynamic states change over time. This means if an attacker switches their strategy, it cannot be detected [1].

Recent research, to move machine learning models away from back-end error correction to active front end defensive strategies, has explored zero-cost algebraic localisation techniques which track an attacker's position along a fibre optic link [7]. This was done by inverting the sifting-phase QBER equations, which maps the asymmetric channel noise on either side of an active tap to approximate the attackers distance without needing large computational power [7]. Even though this method introduces a cost-effective and scalable approach, it introduces a critical problem known as the central vanishing gradient. This happens because optical error propagation across a uniform fiber line is symmetrical, which means the algebraic derivative flattens at the exact midpoint of the communication link. This allows an attacker to hide within this point as the QBER naturally triggers unstable fluctuation which causes large prediction errors.

BB84 implementations are vulnerable to different attack strategies beyond the intercept-resend model used in this project. Photon-number-splitting (PNS) attacks involve skimming one of two photons sent down the quantum channel when the sender's pulse contains multiple photons, allowing the remaining signal to go undisturbed [25]. Detector-blinding attacks involve shining intense light into the single-photon detectors, forcing them to stop registering normal single-photon detection, allowing the attacker to selectively trigger controlled responses without raising suspicion [18]. Trojan-Horse attacks involve injecting light directly into the sender and receivers equipment to infer their internal state settings, attacking the hardware rather than the protocol itself [8].

This research focuses on the intercept-resend attack model for two reasons: firstly, the intercept-resend attack produces observable changes to the channel statistics simultaneously, such as QBER, timing jitter, and dark-counts, making it the appropriate choice to evaluate a multi-modal detector, as opposed to the above attack methods, such as detector-blinding that targets a single hardware side-channel. Secondly, existing literature has already developed countermeasures for the other attack types mention; the decoy-state method for PNS attacks[17] and measurement-device-independent QKD for detector-blinding and other detector side-channel attacks [16], but neither addresses an adaptive eavesdropper who suppresses their QBER signature during an intercept-resend attack whilst leaving trace information in other channel statistics. Classic implementations of intercept-resend attack detection models only account for static QBER [7], allowing a direct comparison to whether a multi-modal, temporally-aware model can outperform prior implementations on the same quantum telemetry dataset.

The studies above each leave a specific gap that motivates the methodological choices made in Chapter 3. Dasari et al. treat phase visibility and QBER as static variables, which do not capture an eavesdropper whose signature intensity varies between episodes or channels; this motivates an adaptive strategy where the attacker implements a variable intensity and channel suppression attack model used in this work.

Fiorini et al.'s localisation approach relies only on QBER and therefore fails where the derivative vanishes, meaning if a detector is built only to monitor one signature, there is an inherent structural blind spot. This motivated the use of multiple physical channels together in our methodology (Section 3.1) rather than a single observable statistic, so if one signal is weak then potentially an attack can be detected through a compensation of trace information in another.

Jha et al.'s Gaussian process regression achieves strong sample efficiency, but fails to generalise when the channel conditions are dynamic, which motivates the use of a temporally aware architecture that models how the channel evolves over a window of time, rather than a model that does not factor in the history of the channel data.

Chapter 3

Methodology

This chapter outlines the proposed framework for passive anomaly detection in QKD networks, detailing the telemetry simulation, data preprocessing, neural network architecture, and baseline benchmarking strategies.

3.1 Synthetic QKD Telemetry Generation

To evaluate the performance of the anomaly detection framework, a realistic simulation environment was created to model the physical telemetry of a QKD link that uses the BB84 protocol. In QKD implementations, the quantum channel is susceptible to environmental and hardware noise which must be accurately portrayed in a simulation to train a robust machine learning classifier. The simulation generates 10,000 timesteps representing multi-modal features: Quantum Bit Error Rate (QBER), Single-Photon Avalanche Diode (SPAD) timing jitter and dark count rates.

The dataset models a point-to-point architecture where an eavesdropper (Eve) attempts an intercept-resend attack during a random set of timestep windows, or attack windows. Twenty-four attack windows are generated per run, each with a randomly generated start time, a duration drawn uniformly from 60 to 200 timesteps, and a randomised minimum gap of 120 to 360 timesteps between the end of the last attack window and the start of a new one. These parameters were chosen to reflect a real adaptive eavesdroppers constraint. The windows are long enough for the eavesdropper to gain useful information, but bounded enough to avoid sustained QBER rise in the channel that could be caught by threshold monitoring. Finally, we included a limited amount of attacks (24), to mimic real channel data, whilst still giving enough attack information to the models. This produces an attack window in approximately 26% of the total timesteps, producing behaviour similar to that of a realistic sparse eavesdropping activity timeline. This intercept-resend attack inherently alters the distribution of the physical telemetry, making it possible to train a model to analyse the telemetry and predict whether an eavesdropper was present during transmission from sender

to receiver.

Additionally, in each episode a random intensity value $\iota \sim \mathcal{U}(0.05, 1.0)$ is assigned, representing the intensity of channel telemetry disturbance. Instead of injecting a uniform attack with maximum intensity for the full duration, each attack disturbance is scaled between a random maximum intensity ι and normal background noise. This way there is a large range of attack intensity, from an almost imperceptible $\iota = 0.05$ to the maximum intensity of $\iota = 1.0$. This reflects how practical eavesdroppers will vary in equipment and quality. Furthermore, 40% of the episodes are allocated as stealthy attacks. During these attacks the QBER channel's intensity value is scaled between 15% and 40% of the episode's chosen intensity value, timing jitter is moderately suppressed between 50% and 70% intensity, and dark count remains unscaled at full attack intensity. This models an adaptive eavesdropper strategy that tries to suppress its channel QBER signature, which is the most commonly monitored statistic by single-observer detectors. (This distinction between standard and stealthy episodes is the mechanism this dissertation uses to evaluate H2).

3.1.1 Quantum Bit Error Rate (QBER) Modelling

During normal operation, thermal and magnetic fluctuations, decoherence of qubits and channel loss can influence the state of the qubit sent. QBER is the ratio of incorrect bits to the total number of transmitted bits. This is modelled with a Gaussian-perturbed baseline:

$$\text{QBER}_{\text{baseline}} = 0.015 + \mathcal{N}(0, 0.012) \quad (3.1)$$

When Eve intercepts and resends the qubit using an incorrect basis, it inherently changes the quantum state, increasing the QBER [2]. Under theoretical conditions this introduces a 25% error rate as concluded by [2]. In the simulation the attack causes the QBER mean to increase according to:

$$\text{QBER}_{\text{mean}} = 0.015 + \iota \cdot 0.055, \quad \iota \in [0.05, 1.0] \quad (3.2)$$

with probabilities bounded within $[0, 1]$.

3.1.2 SPAD Timing Jitter

Avalanche photodiodes (APDs) and Single-Photon avalanche diodes (SPADs) used in quantum receivers show uncertainties between photon absorption and the resulting electrical avalanche, known as timing jitter [22]. The baseline jitter used in the simulation is modelled as:

$$J_{\text{baseline}} \sim \mathcal{N}(\mu = 0.0 \text{ ns}, \sigma = 0.1 \text{ ns}) \quad (3.3)$$

as photons arrive with a small delay either early or late, and with a standard deviation of $\sigma = 0.1$ ns, which is based on the timing jitter figure reported for single-photon avalanche diodes used in QKD receivers [22]. During an intercept-resend attack, Eve introduces temporal delay due to the processing load of measurement and photon regeneration, which shifts the distribution mean during attack window:

$$\mu_{\text{jitter}} = \iota \cdot (-0.55), \quad \iota \in [0.05, 1.0] \quad (3.4)$$

representing a shift of up to -0.55 ns at full intensity.

3.1.3 Dark Count Rates

Single-photon detectors sometimes register false positives which can be triggered by thermal and environmental noise [22], or weak coherent sources. Because both occur as discrete independent events the simulation can model this using a Poisson distribution [21], with baseline rate:

$$D_{\text{baseline}} \sim \text{Poisson}(\lambda = 0.05) \quad (3.5)$$

Eve’s active probing and injection of photons can increase the thermal noise at the receiver, leading to an increase in the rate parameter during attack windows:

$$\lambda_{\text{attack}} = 0.05 + \iota \cdot 0.11, \quad \iota \in [0.05, 1.0] \quad (3.6)$$

up to a maximum rate parameter increase of 0.11 at full intensity.

By combining variable-intensity, sparsely distributed and stealthy attack episodes into the data generation, the resulting dataset avoids predictable separability of attacks, with a fixed magnitude signature that better encapsulates a realistic test environment. This allows for a better evaluation of whether the LSTM generalises across all attacker behaviour, including behaviour meant to evade single-observer detection.

Figure 3.1 shows a representative simulated run of all 10,000 timesteps, showing that elevated QBER and shifted jitter correlate with the attack ground-truth labels in the bottom panel, whilst dark-counts remain unaffected outside of occasional baseline spikes consistent with the Poisson noise model.

3.2 Data Preprocessing and Sequence Formatting

The physical telemetry data created from the quantum channel outputs a continuous multivariate time-series. For the LSTM model to process this data it must be preprocessed to preserve structural proportions of the data as well as keeping chronological dependencies.

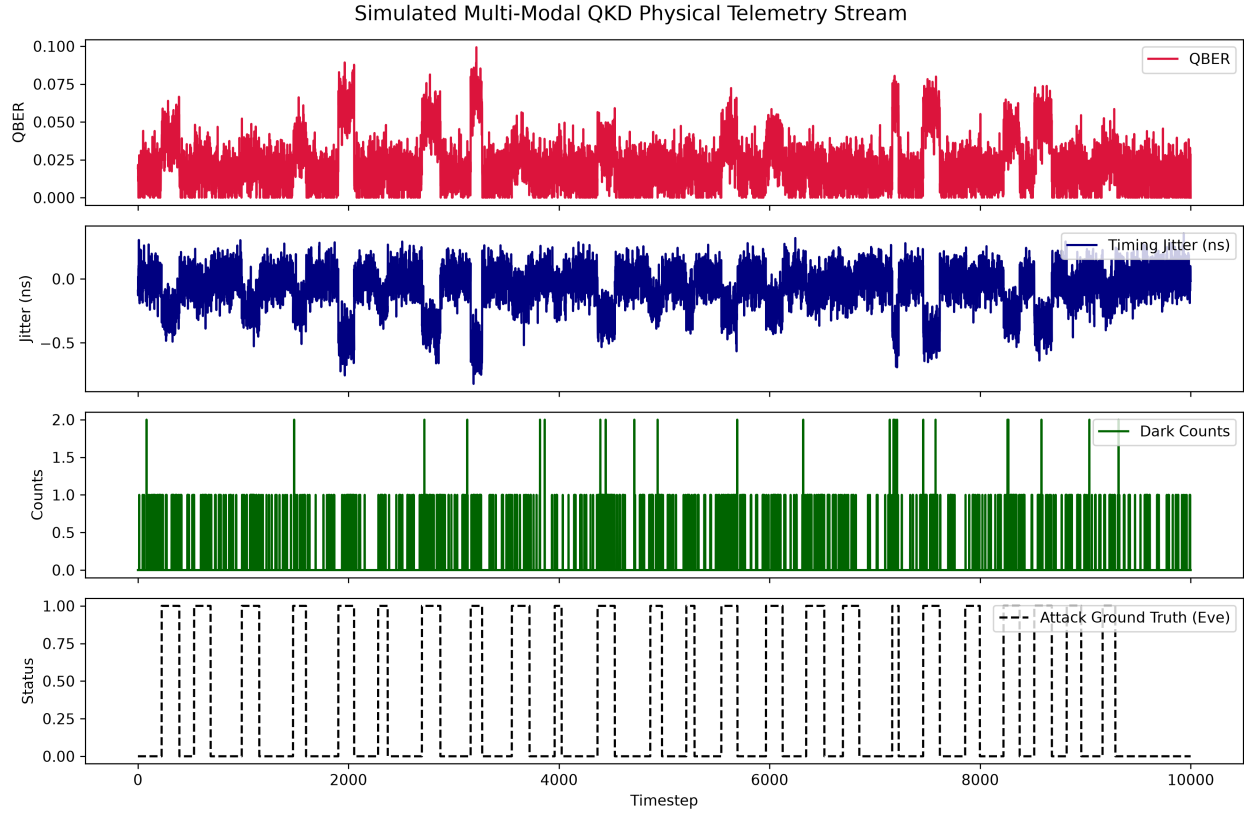


Figure 3.1: Simulated multi-modal QKD telemetry stream over 10,000 timesteps, showing QBER, timing jitter, dark counts, and attack ground truth.

3.2.1 Data Standardisation and Leakage Prevention

As the LSTM model relies on gradient descent optimisation, it is highly sensitive to the magnitude of the input features. To ensure that features with higher ranges, such as dark counts, do not dominate the loss gradient, Z-score standardisation is utilised. Each feature f is transformed to have a mean of zero and a unit variance.

$$z = \frac{f - \mu}{\sigma} \quad (3.7)$$

μ and σ are calculated only on the chronologically split training set and used to scale the test set to ensure the model's evaluation can handle unseen quantum telemetry data. This method also maintains strict experimental integrity and minimises data leakage.

3.2.2 Temporal Windowing Strategy

Classical machine learning algorithms process data as independent identically distributed samples, which does not keep sequence order. However, eavesdropping tactics such as Photon-Number Splitting attacks or intercept-resend attack introduce gradual and sustained statistical changes in the quantum channel as opposed to fast isolated spikes. To capture the temporal information of the channel the data is transformed using a sliding lookback window.

With the standardised dataset represented as a 2D matrix $X \in \mathbb{R}^{N \times F}$ where N is the number of timestamps, F is the number of physical features (in this case $F = 3$) and the sliding window with size $W = 50$ time steps is passed over the timeline with a step of 1. Then for a given timestep t , the extracted sequence S_t is defined as:

$$S_t = [x_{t-W+1}, x_{t-W+2}, \dots, x_t] \quad (3.8)$$

This transforms the dataset from a 2D matrix into a 3D tensor with dimensions (S, W, F) where $S = N - W + 1$ is the number of windows created. Ground truth labels $y_t \in \{0, 1\}$ are assigned based on the last timestep of the window. This way the LSTM network has been provided the previous 50 timesteps allowing it to model and distinguish the timesteps of normal channel noise against third-party interference.

3.3 LSTM-Based Anomaly Detection Framework

Logistic Regression and Random Forest classifiers treat each input feature as an independent sample of the data without keeping information about order. In the case of an intercept-resend attack, there is a sustained and evolving statistical shift in QBER, timing jitter, and dark-count rate as opposed to isolated spikes in the channel. Differentiating between these

statistical shifts and normal channel noise requires modelling how the signal changes over a window of time.

A Long Short-term Memory (LSTM) [10] was selected as it is optimal for dealing with the nature of the channel and was chosen over three plausible alternatives for reasons specific to this task, detailed below. Compared to other models that focus on isolated spikes, such as Transformers, LSTMs require substantially less training data to generalise [30], a relevant constraint as real QKD hardware telemetry is scarce, expensive and slow to collect. Transformers typically need more training examples to learn useful patterns than the roughly 8000 training windows implemented in this project, making it a worse fit for this data constrained environment, despite being theoretically advantageous, as they can model dependencies on the entire lookback window directly, rather than sequentially as an LSTM does. However, this usually requires more than 8000 training windows. In addition, relative to 1D convolutional networks that are effective when detecting local, predictable patterns, an LSTM’s recurrent cell state is better equipped when dealing with variable length, gradually changing channel statistics, because an attack’s duration is not fixed or consistent. A convolutional network would need to be tuned to a specific attack duration, whereas the LSTM’s recurrent state can, in principle, track a disturbance regardless of how long it persists in the lookback window. Finally, compared to the lighter and faster gated-recurrent-unit (GRU) network, the LSTM has a higher capacity for retaining temporal information across the 50-timestep lookback window used in this work, at the cost of additional parameters and slightly slower training. This choice does come with a marginal additional computation cost that is justified by the importance of producing a real-time classification decision.

This temporally aware design is also what directly targets the limitations of the single observable channel implementations shared by Fiorini et al. and Dasari et al. By modelling three channels over time rather than one channel at a single instant, the architecture permits our detector to capture the varying trace information over all channels, that could plausibly show as regular channel noise to a single-channel detector.

3.3.1 Recurrent Architecture

The LSTM takes the 3D sequence tensors as an input, where each sequence is defined in (3.8), where the sequence length is 50 and $x_t \in \mathbb{R}^3$ represents the feature vector, made up of QBER, timing jitter and dark counts at timestep t . At each timestep, the LSTM cell updates its hidden state h_t and cell state c_t using three gating mechanisms that decide what information is forgotten, added, and output which is governed by these formulas:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (3.9)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (3.10)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (3.11)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (3.12)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (3.13)$$

$$h_t = o_t \odot \tanh(c_t) \quad (3.14)$$

where f_t , i_t and o_t are the forget, input, and output gates respectively, $\tilde{c}_t$ is the candidate cell state, $\sigma(\cdot)$ is the sigmoid function, $\odot$ is element-wise multiplication, and $[h_{t-1}, x_t]$ is a concatenation of the previous hidden state with the current input. The forget gate f_t (3.9) is responsible for the behaviour discussed in Section 5.2, and allows the network to keep weak trace information carried in cell state c_t across timesteps even when the strongest signal is suppressed, rather than being forced to average or discard it.

Two LSTM layers are stacked with a hidden state dimension of $h = 64$ nodes, which was chosen as a standard default but is not tuned specifically for this dataset. After each timestep the hidden state h_t and the cell state c_t are updated using the current input x_t , with the previous hidden state h_{t-1} , and previous cell state c_{t-1} , as can be seen in (3.9) to (3.14). Using multiple layers lets the network learn complex physical behaviours shown in the quantum channel. However, a measure must be taken to prevent over-fitting on the training data; in this case dropout regularisation is used between the LSTM layers with a probability of 0.2.

3.3.2 Classification Head

To decide a classification for each sequence, only the hidden state of the last timestep of a window is extracted, h_W . This hidden state vector contains the network's entire understanding of the timestep window, including the temporal information, which is then passed through a linear layer with a weight matrix $W_c \in \mathbb{R}^{1 \times 64}$ and bias term b_c . Finally, a sigmoid activation function is applied to ensure the output has a strict probability bound of $[0, 1]$. The anomaly probability is calculated using:

$$\hat{y} = \sigma(W_c h_W + b_c) = \frac{1}{1 + e^{-(W_c h_W + b_c)}} \quad (3.15)$$

with $\hat{y} \geq 0.5$ signifying the presence of an attack on the quantum channel.

3.4 Classical Machine Learning Baselines

To justify the computational workload of using the LSTM model, its performance was benchmarked against two standard machine learning models. Logistic Regression [5] and Random Forest [4] were chosen to compare their performance against the proposed LSTM and to represent well-established non-sequential baseline models.

3.4.1 Sequence Flattening and Loss of Temporal Topology

Classical models expect the input in the form of 2D matrices where each feature is treated as an independent variable. Therefore, the 3D tensors generated must be flattened into 2D vectors before use.

$$X_{flat} \in \mathbb{R}^{S \times (W \cdot F)} \quad (3.16)$$

Each window and each feature is collapsed into a single flat vector made up of 150 variables ($W = 50$ and $F = 3$). This allows for both the Logistic Regression and Random Forest to process the same data used by the LSTM model, however through flattening the data the temporal information is lost.

3.4.2 Benchmark Objective

The classical models predict the presence of an eavesdropper based on the aggregate statistical variation in the 150 variable vector, ignoring the order in which the variations occurred. Comparing these models with the LSTM, which includes temporal information, allows us to quantify the value of sequential memory when trying to detect the signature of a sustained intercept-resend attack.

3.4.3 Cross-Dataset Evaluation

The fixed dataset evaluation above introduces no variance in the dataset, but rather introduces variance in model initialisation and training procedure. This leads us to require more generalised evaluation of the model, to see if the observed advantages hold true beyond the specific dataset instance. To address this, an additional evaluation was implemented where each model is trained once on a pool of ten concatenated and independently generated telemetry datasets after being windowed separately, so no lookback window passes between two adjoining datasets, preventing test set leakage. Apart from this change, the evaluation is kept consistent with the fixed-dataset parameters. Following the training, each model is then evaluated separately on eight additional unseen datasets, each generated independently using the same process described in Section 3.1, producing eight independent paired measurements per model, which allows for paired significance testing. Finally, aggregate ROC-AUC and

stratified stealthy-attack recall are recorded for each test dataset, allowing further evaluation of H1 and H2 under a test of generalisation.

Chapter 4

Experimental Setup and Simulation

The simulation of the quantum channel telemetry and the BB84 intercept-resend attack was developed using Python, and various libraries used to manipulate the data such as Pandas and NumPy. The LSTM eavesdropper detection framework was implemented with the PyTorch deep learning library. For the classic baseline models (Random Forest and Logistic Regression), the Scikit-Learn library was used.

4.1 Training and Optimisation Hyper-parameters

For the fixed-dataset evaluation, the dataset, which contained 10,000 timestamps, was split into an 80/20% training and test set respectively. For the cross-dataset evaluation, ten datasets are pooled together for training and the models are evaluated separately on eight unseen datasets. To minimise classification error, a Binary Cross-Entropy (BCE) function was applied to the LSTM which awards a larger penalty when the model's prediction probabilities were significantly different from the ground truth labels. For N sequences where $y_i \in \{0, 1\}$ is a true label and $\hat{y}_i$ is the probability predicted by the model, the BCE loss is calculated as:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (4.1)$$

To optimise the model, an Adaptive Moment Estimation (Adam) [12] optimiser was used, as it performs well with noisy data and sparse gradients. The model was trained over 15 epochs with a learning rate of 0.001 and batch sizes of 64 sequences. For the Random Forest classifier 100 estimators were used, and for the Logistic Regression model a maximum of 1000 iterations were used to ensure that the model converged.

The majority of hyper-parameters used for training were set to standard defaults or supported by recent literature, such as the Adam optimiser's learning rate following

the original paper’s recommendation [12], rather than being obtained through extensive parameter testing. The Adam optimiser’s learning rate and the Random Forest estimators value both were based on default configurations for these methods, chosen as standard starting values proven by previous research to generalise well to most problems, rather than requiring tuning specific to this dataset. The number of training epochs was informed by the training loss curve seen in Figure 4.1, which shows the binary cross-entropy loss converging and stabilising within this range, with no evidence of divergence or loss increasing which would show over-fitting. A full hyperparameter sweep was considered for this project including analysis of learning rate, batch size, hidden dimension and dataset size, but was considered to be outside the project scope given time constraints and is identified as a point for future work in Chapter 7.

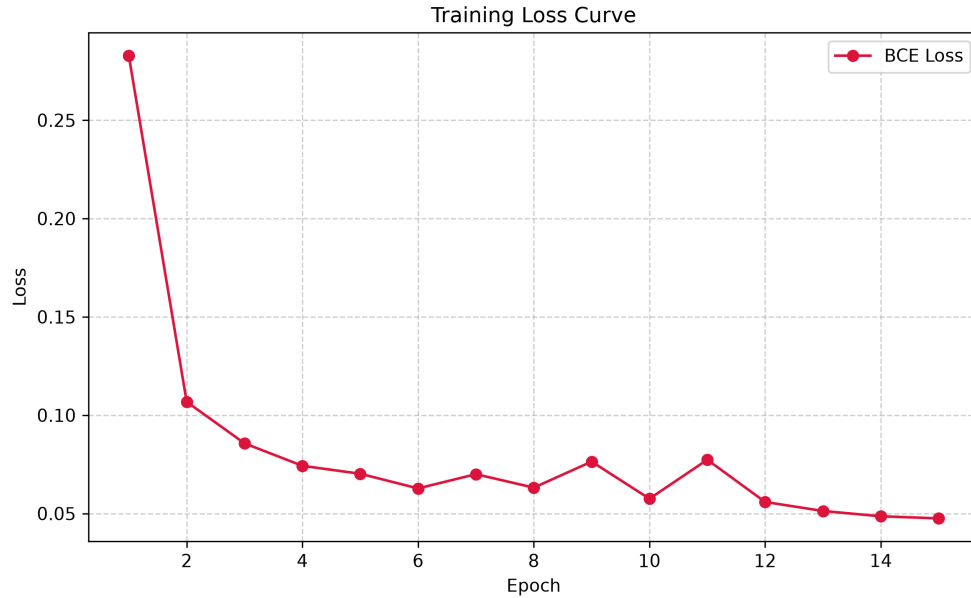


Figure 4.1: LSTM training loss (binary cross-entropy) over 15 epochs, seed 42, showing stable convergence.

4.2 Evaluation Metrics

To assess and compare the models’ classification capabilities on the dataset, several metrics were calculated to compare performance: Accuracy, F_1 -score, and area under the receiver operating characteristic curve (ROC-AUC). The accuracy, F_1 -score, precision and Recall metrics were calculated as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.2)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (4.3)$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN} \quad (4.4)$$

where TP represents correctly identified attacks, TN represents correctly identified normal operation on the channel, FP represents falsely identified attacks, and FN represents attacks that went undetected. F_1 -score was used in addition to accuracy as it provides a deeper evaluation involving precision and recall. Finally, ROC-AUC evaluates the model's ability to distinguish between normal behaviour and attack behaviour on the channel, where a score of 1.0 represents perfect identification and 0.5 indicates random guessing. The ROC-AUC score shows how reliable the models are over varying network sensitivity tolerances.

Recall is particularly important to this research, as it measures the proportion of attacks that were correctly identified, showing how many eavesdropping instances the model would miss. This makes it a primary metric when evaluating detection performance separately for standard and stealthy attacks.

To test whether the LSTM's advantage over each baseline is statistically significant, a paired t -test is used. For n matched pairs (each pair matched by either the same seed or same unseen dataset), with per-pair differences $d_i = x_{i,\text{LSTM}} - x_{i,\text{baseline}}$, the test statistic is calculated using the formula:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \quad (4.5)$$

where $\bar{d}$ is the mean of the paired differences and s_d is their standard deviation. Assuming the LSTM does not outperform the baseline models (null hypothesis) we derive the p -values that show the probability of observing a mean difference as the one measured, if the null hypothesis were true, with $p < 0.05$ signifying the null hypothesis is rejected and there is a significant statistical advantage for our model. We chose this paired approach to isolate variance to the choice of model alone.

Chapter 5

Results and Discussion

This chapter details the performance of the LSTM gatekeeper against the classical baseline models. To ensure the results are fair for all models in the fixed-dataset evaluation, they were tested on the exact same 80/20 chronological split of the synthetically generated dataset, as well as the results being an average across five distinct training runs using fixed random seeds $\{0,1,2,3,4\}$.

5.1 Aggregate Model Performance

Table 5.1 shows the mean accuracy, F1-score, and ROC-AUC for each model, as well as the standard deviation across the five training seeds.

Table 5.1: Aggregate test-set performance, mean $\pm$ SD across 5 seeds

Model	Accuracy	F1-Score	ROC-AUC
LSTM (proposed)	0.982 ± 0.002	0.970 ± 0.004	0.996 ± 0.000
Random Forest	0.965 ± 0.003	0.938 ± 0.006	0.993 ± 0.000
Logistic Regression	0.974 ± 0.000	0.956 ± 0.000	0.994 ± 0.000

Table 5.1 shows that the LSTM achieved the highest mean value in all three measurements taken: Accuracy, F1-score and ROC-AUC score. Notably, Logistic Regression’s score for standard deviation is zero; this is to be expected as its default solver optimises a loss function to a unique global minimum, which means for each seed, given fixed training data, the standard deviation will always be zero.

The raw ROC-AUC gap between models is relatively small (0.996 for the LSTM versus 0.993-0.994 for both baseline models), which could suggest that the choice between models is negligible. However, the paired seed design, used to create Table 5.2, shows model attribute variance from dataset attribute variance, showing that this advantage, although small, is

consistent in favour of the LSTM. This is why a mean AUC difference as small as +0.0027 against Random Forest still shows as statistically significant ($p=0.0001$), further reinforces our claim through robust and reproducible testing.

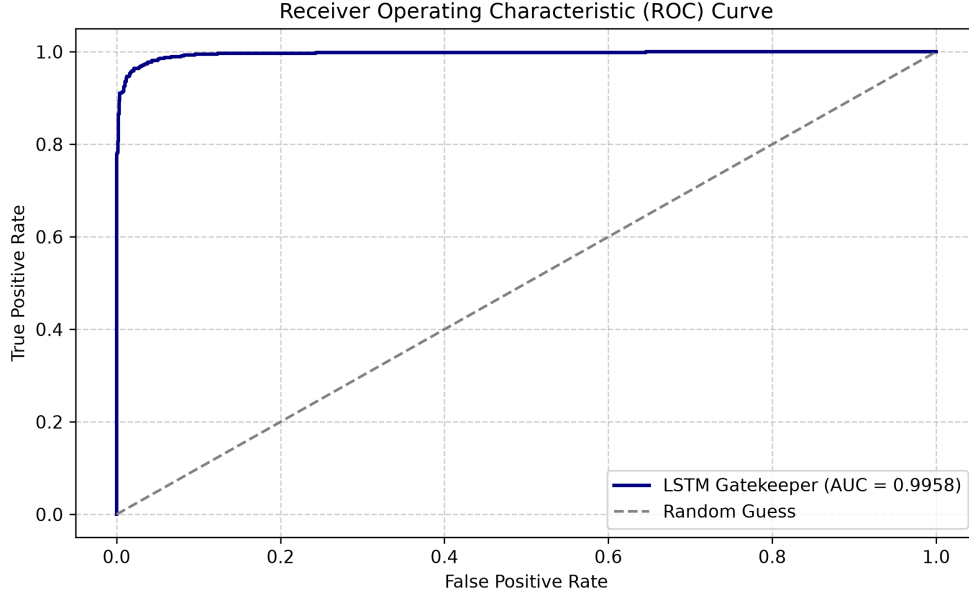


Figure 5.1: Receiver Operating Characteristic curve for the LSTM gatekeeper on the test set (seed 42, AUC = 0.9958).

Table 5.2: Paired t -test results, LSTM vs. each baseline (ROC-AUC, $n = 5$ seeds)

Comparison	Mean Δ AUC	t	p	Significant ($\alpha = 0.05$)
LSTM vs. RF	+0.0027	14.113	0.0001	Yes
LSTM vs. LR	+0.0020	9.445	0.0007	Yes

Comparing both baseline models against the LSTM reveals that its advantage is statistically significant against both Logistic Regression and Random Forest, where $p = 0.0007$ and $p = 0.0001$ respectively. These results provide sufficient evidence to reject the null hypothesis $H_{0,1}$: that the LSTM does not outperform the baseline models in aggregate ROC-AUC and therefore show that our first hypothesis: that a multi-modal, temporally aware classifier outperforms non-sequential baseline models in aggregate, is supported by the fixed-dataset evaluation.

5.2 Stratified Performance: Standard vs. Stealthy Attacks

The test set used contains 579 attack windows, with 138 of those being stealthy episodes, where QBER is heavily suppressed and jitter is moderately suppressed. Table 5.3 shows each model’s recall separately for each attack type.

Table 5.3: Detection recall by attack subtype, mean $\pm$ SD across 5 seeds {0, 1, 2, 3, 4}

Model	Non-stealthy recall	Stealthy recall	Recall drop
LSTM (proposed)	0.990 ± 0.001	0.900 ± 0.019	0.090 ± 0.020
Random Forest	0.984 ± 0.003	0.626 ± 0.026	0.358 ± 0.025
Logistic Regression	0.989 ± 0.000	0.826 ± 0.000	0.163 ± 0.000

All models detect standard attacks at a near perfect rate (0.98–0.99), however performance varies sharply when attacks are stealthy. Random Forest’s recall falls the most, from 0.984 to 0.626, a fall of roughly 35.8%. Logistic Regression’s recall falls by roughly 16.3%, outperforming Random Forest by correctly identifying stealthy attacks roughly 82.6% of the time. The LSTM correctly identifies the higher number of stealthy attacks, only falling 9.0% and correctly identifying 90.0%. These results show that the LSTM outperformed Random Forest by roughly 1.4 $\times$ in stealthy attack recall, as well as being the model where recall dropped the least from standard attacks to stealthy ones.

Table 5.4: Paired t -test results, LSTM vs. each baseline (stealthy recall, $n = 5$ seeds)

Comparison	Mean Δ Recall	t	p	Significant ($\alpha = 0.05$)
LSTM vs. RF	+0.2739	20.683	0.000032	Yes
LSTM vs. LR	+0.0739	8.500	0.001051	Yes

This result reveals what the aggregate finding in Section 5.1 conceals. Most of the attacks are not stealthy attacks, 138 stealthy out of 579 total attacks, and all three models detect this type of attack almost perfectly, compressing the overall ROC-AUC score toward the top of its range. As a result, the models’ aggregate scores cluster closely together, while the LSTM’s advantage over the baseline models emerges through the detection of stealthy attacks. The LSTM achieves a recall 27.4% higher than Random Forest and 7.4% higher than Logistic Regression, with a degradation of recall 4.0 $\times$ smaller and 1.8 $\times$ smaller respectively between the attack types. This supports the theory that a multi-modal, temporally aware classifier has an advantage over a non-sequential baseline model, and that the advantage is largest against an adaptive eavesdropper who suppresses their signature across channels,

rather than uniformly across all attack conditions. This shows that the LSTM’s advantage is statistically robust as the paired t-tests in Table 5.4 confirm the LSTM’s stealthy attack recall advantage is significant against both Random Forest ($p = 0.000032$) and Logistic Regression ($p = 0.001051$).

When an attack is designed to be stealthy, the QBER is heavily suppressed while the jitter and dark counts are relatively unchanged, a model that flattens the timestep window of size 50 into a single 150 sized variable vector, such as Random Forest and Logistic Regression must, treats every time step equally regardless of when it occurred. This means that if there is a brief or gradually building disturbance in the channel signatures, they can be diluted by the surrounding signatures of normal readings and go undetected.

In comparison, the LSTM’s recurrent state is able to remember weak trace information across the window rather than averaging all information, which could explain why its recall falls the least when the strongest channel signature (QBER) is suppressed. The LSTM does not rely on one singular channel signature in isolation or singular timestep to make its classification decision.

This research does not test this mechanism directly, although this could be done through attention weight inspection, but this is offered as the most consistent explanation of the observed pattern. This is presented more precisely as a direction for future work in Chapter 7.

This also highlights that in an operational QKD deployment, eavesdropper detection is not indicative of detection success, as a single undetected intercept-resend attack compromises the security behind every key bit exchanged during that attack window, regardless if the detector managed to find all 579 attack windows in the test set. Compared to the LSTM, the baseline models have a stealthy attack recall of 62.6% for Random Forest, which accounts for just over a third of stealthy intercept attempts going undetected and 82.6% for Logistic Regression which is roughly one in six. In these undetected attacks the key material is treated as secure when it is not. The LSTM achieving 90.0% is a small reduction in undetected stealthy attacks is comparatively significant to both baselines, with Random Forest’s miss rate on stealthy attacks standing at 37.4%, even though the aggregate ROC-AUC only differing by a few thousandths between all three models.

This is the central argument for the use of stratified evaluation referenced above, as selecting a gatekeeper model on aggregate ROC-AUC alone would present as having very little difference in performance, despite the fact that they differ by a factor of nearly four in how often they fail against the exact strategy an adaptive adversary is likely to employ.

5.3 Cross-Dataset Evaluation

To assess whether the findings in Sections 5.1 and 5.2 generalise beyond the single dataset instance, each model was additionally trained on a pool of ten datasets and evaluated separately on a further eight unseen datasets, following the protocol described in Section 3.4.3.

5.3.1 Aggregate Cross-Dataset Performance

Table 5.5 shows the mean accuracy, F1-score and ROC-AUC for each model across the eight unseen test datasets.

Table 5.5: Cross-dataset aggregate test-set performance, mean $\pm$ SD across 8 held-out datasets

Model	Accuracy	F1-Score	ROC-AUC
LSTM (proposed)	0.966 ± 0.014	0.944 ± 0.026	0.982 ± 0.013
Random Forest	0.955 ± 0.015	0.923 ± 0.026	0.982 ± 0.011
Logistic Regression	0.952 ± 0.015	0.920 ± 0.025	0.979 ± 0.012

This table shows that, unlike the fixed-dataset evaluation, the mean ROC-AUC is nominally higher for Random Forest (0.982), though the LSTM achieved an almost identical mean ROC-AUC of 0.982 to three decimal places. Table 5.6 shows that we fail to reject the null hypothesis of no difference between the LSTM and Random Forest across the eight unseen datasets ($p = 0.9274$), but it retains its advantage over Logistic Regression ($p = 0.0107$) in this more generalised evaluation.

Table 5.6: Paired t -test results, LSTM vs. each baseline (cross-dataset ROC-AUC, $n = 8$ held-out datasets)

Comparison	Mean Δ AUC	t	p	Significant ($\alpha = 0.05$)
LSTM vs. RF	-0.0002	-0.094	0.9274	No
LSTM vs. LR	$+0.0032$	3.447	0.0107	Yes

The similarity in results is not just a matter of insufficient data to prove a difference. When examining the performance of each dataset individually, the two models alternate between which one has better results: the LSTM performs better in six out of the eight unseen test sets and Random Forest performs better on the remaining two. However, Random Forest’s margin of victory on those two datasets is greater than LSTM’s margin on all six of its victories, which explains the non-significant aggregate results.

The lack of a significant aggregate advantage over Random Forest may partly be influenced by the absence of parameter optimisation performed on the LSTM, while the parameters chosen for Random Forest are standard defaults, that are a competitively robust choice validated across a wide range of problems.

5.3.2 Stratified Cross-Dataset Performance

Table 5.7 shows each model’s recall for standard and stealthy attacks, averaged across the eight held-out datasets.

Table 5.7: Cross-dataset detection recall by attack subtype, mean $\pm$ SD across 8 held-out datasets

Model	Non-stealthy recall	Stealthy recall	Recall drop
LSTM (proposed)	0.969 ± 0.026	0.888 ± 0.105	0.082 ± 0.120
Random Forest	0.939 ± 0.048	0.772 ± 0.179	0.167 ± 0.217
Logistic Regression	0.942 ± 0.041	0.806 ± 0.158	0.136 ± 0.189

As shown in Table 5.7, the LSTM’s stealthy recall (88.8%) remains the highest of the three models, and its recall drop between attack subtypes (8.2%) is the smallest, compared to both Random Forest (77.2% stealthy recall) and Logistic Regression (80.6% stealthy recall) dropping by 16.7% and 13.6% respectively.

Table 5.8: Paired t -test results, LSTM vs. each baseline (cross-dataset stealthy recall, $n = 8$ held-out datasets)

Comparison	Mean Δ Recall	t	p	Significant ($\alpha = 0.05$)
LSTM vs. RF	+0.1155	3.525	0.0097	Yes
LSTM vs. LR	+0.0821	3.446	0.0108	Yes

Table 5.8 confirms both comparisons are statistically significant with $p = 0.0097$ against Random Forest and $p = 0.0108$ against Logistic Regression, so we reject the null hypothesis that the LSTM does not outperform the baseline models. Noticeably, the standard deviations observed are larger than in the fixed-dataset evaluation, reflecting the variance in dataset composition and attack intensities across the eight test sets.

Chapter 6

Discussion

The results in Chapter 5 show that the hypothesis that a multi-modal, temporally aware classifier outperforms non-sequential baseline models is supported by the primary fixed-dataset evaluation. The LSTM’s advantage over Logistic Regression and Random Forest is statistically significant ($p = 0.0007$ and $p = 0.0001$ respectively), when evaluating the aggregate ROC-AUC score. This would suggest that in practical implementations there is a benefit of using a temporally aware model on this fixed dataset, though its general advantages are examined below.

The LSTM has a modest advantage in aggregate score, but this can undervalue the LSTM’s advantage. Because 76% of attacks in the evaluation are set at standard intensity values and have high detection rates, all models are capable in this scenario. The true advantage of the LSTM is concentrated when observing its ability to detect stealthy attacks, which is the most likely to be the strategy utilised by a real adaptive eavesdropper. This makes our second hypothesis strongly supported. The LSTM’s recall for stealthy attacks (90.0%) exceeds both Logistic Regression (82.6%) and Random Forest (62.6%), as well as decreasing the least when evaluating between standard intensity and stealthy attacks, only dropping by 9.0% compared to 35.8% and 16.3% for Random Forest and Logistic Regression.

This highlights the importance of how security systems are tested. Overall accuracy or ROC-AUC alone are inadequate metrics for evaluating eavesdropper detectors intended for real world application, because it treats all attack types with the same weighting, based on presence on the channel instead of their operational threat. A model that performs well to detect normal attacks, but efficiency significantly reduces when attempting to detect sophisticated adaptive attacks can still achieve a high aggregate score, whilst being a weaker choice in practice. Stratified evaluation by attack type used in this work is recommended as standard practice for future QKD anomaly-detection research.

6.1 Generalisation Beyond a Fixed Dataset

Results seen in Sections 5.1 and 5.2, where all three models were trained and tested on the same dataset, isolate variance from model initialisation and training from variance due to the dataset itself, but do not test whether the dataset itself is particularly favourable or unfavourable to any model. To test this, each model was additionally trained on a pooled set of ten datasets and tested separately on eight further unseen datasets as seen in Section 5.3. In this evaluation, the LSTM’s aggregate ROC-AUC advantage is no longer statistically significant over Random Forest ($p = 0.9274$). However, the LSTM remains statistically advantageous over Logistic Regression in this more generalised evaluation ($p = 0.0107$).

The key difference can be seen in the stratified evaluation on stealthy attack detection. The LSTM achieved a stealthy-attack recall of 88.8% across the eight unseen datasets; this is substantially higher than both Random Forest (77.2%, $p = 0.0097$) and Logistic Regression (80.6%, $p = 0.0108$), with recall drop continuing to be the smallest out of all three models (8.2%), compared to 16.7% for Random Forest and 13.6% for Logistic Regression. These results are similar to those achieved in the fixed-dataset evaluation in Section 5.2 and show that H2’s claim remains consistent regardless of which datasets the models are trained and tested on, unlike H1. Additionally, larger standard deviations can be seen compared to fixed-dataset evaluation, with the LSTM stealthy recall being observed as 0.105 in the cross-dataset evaluation in Table 5.7, in comparison to 0.019 in the fixed-dataset evaluation in Table 5.3. This reflects genuine attack composition and intensity variation across each dataset, which strengthens confidence that the advantage is a property of the model rather than a favourable dataset.

These results highlight that aggregate performance metrics are unreliable by themselves for selecting a gatekeeper model and they should be paired with stratified evaluation, since small aggregate differences can be hidden once a model is exposed to an unseen dataset. The LSTM’s advantage against an adaptive attacker who suppresses their signature is a reliable finding of this research as it survives a more strict generalisation test.

6.2 Validity of the Approach

This research creates a synthetic dataset, but its validity can be questioned as early iterations of the dataset were found to be invalid for its intended purpose during development.

The first iteration of the telemetry dataset injected two distinct, fixed magnitude attacks across two static windows. This produced strong results, with all three models detecting attacks at a near perfect rate. A single-threshold classifier could achieve 100% accuracy on this dataset, and all 3 models achieved ROC-AUC scores of borderline 1.0. We recognised that the dataset was linearly separable meaning any threshold-based classifier could achieve near perfect performance. Our decision was to redesign the dataset generator to deliberately introduce a more challenging, adaptive attack behaviour that would support our second

hypothesis and test the models under conditions closer to a real adversary. For our next iteration, attacks were of variable intensities, sparsely separated across episodes, as well as a stealthy attack subtype that suppresses its signatures across channels. These changes allowed for direct stratified evaluation. This is stated to show that the difficulty of the dataset is a deliberate decision made to test models under realistic challenging conditions, rather than an arbitrary or convenient set of noise parameters.

Another important validity consideration in this project is reproducibility. To address this, all random number generation used in dataset generation, model weight initialisation, and data loader shuffling was explicitly seeded, and results pertaining to averages were taken across five seeds with the same dataset used for each seed. This seeding allowed us to evaluate the models multiple times on the same dataset and apply paired significance testing so the models' behaviour could be compared rather than relying on a single potentially favourable run. Reproducibility and robustness were the main influences for the cross-dataset evaluation, which measures the model's ability to generalise to unseen data rather than relying on the results of one potentially favourable dataset.

A further validity concern to consider is the accuracy of the simulated telemetry data itself compared to real QKD hardware. The baseline value for QBER (1.5%) was chosen to be broadly consistent with published figures [29]. However, real single-photon detectors used in QKD systems show dark count and timing jitter characteristics that vary by several orders of magnitude depending on the detector technology used [26]. The jitter standard deviation used in this simulation (Section 3.2) was chosen to loosely match reported SPAD timing-jitter figures [22], as jitter is expressed in nanoseconds. This does not extend to the value of our dark-count rate, which is expressed as a poisson parameter per simulated timestep, rather than a real unit of time; because the timestep used in this research is not based on any real duration of time, this value cannot be meaningfully compared to a hardware dark-count rate measured in Hz. These parameter values should not be treated as representative of a calibrated practical benchmark, but of how quantum channels behave in principle.

6.3 Scope and Underpinning Assumptions

Initial discussions of this project considered a broader treatment of quantum algorithmic threats, including coverage of Shor's and Grover's algorithms and AI-driven protocol optimisation within the Bob-Alice model. Before implementation began, the scope was narrowed to focus on a multi-modal detection model targeting intercept-resend attacks specifically. We did this to address the gap in literature, that does not explore instances where attackers use adaptive stealthy-attack strategies. This focus allowed us to improve QKD gatekeepers' ability to detect signature suppressing attacks, which addresses a key limitation of single-observable detectors.

Real QKD hardware telemetry data is expensive and slow to collect, which led us to create our synthetic dataset, with a varied range of attack intensities and stealth levels, which may

not have been possible with a small set of real data samples. However, we acknowledge this as a limitation because our specific noise parameters, attack magnitudes, and the stealth intensity ranges are not physically justified. Our findings that the LSTM outperforms the baseline models substantially whilst identifying stealthy attacks are expected to be robust to moderate changes in these parameters, as the structural properties of the models are taken into consideration, rather than specific numerical tuning. The precise results stated in this project should be treated as specific to this simulation rather than as a generalised constant.

6.4 Limitations

Three limitations should be noted. First, this project only addresses the intercept-resend attack model, with photon-number-splitting, Trojan-Horse and detector-blinding attacks not being addressed. These were identified in the literature review, but our implementation does not evaluate the model against these attacks and therefore should be used to complement approaches to address other attack types. Secondly, the stealthy attack model used suppresses three signature intensities uniformly at random across episodes, which does not model an eavesdropper who could potentially adapt their strategy in response to the detector's feedback, which would represent a threat model that is stronger and more realistic than the model used in this project. This, however, could be the scope for future work. Another key limitation was the lack of parameter optimisation for the models which could statistically compare all models performing at their optimal rate, which would show a distinct performance difference. Finally, for our first evaluation, using a fixed-dataset, we trained the model multiple times on the same data to ensure results were consistent; however, for the cross-dataset evaluation each model was trained only once on the pooled training data rather than once per training dataset. This does not account for training-run variance and therefore; a more complete evaluation would additionally train across several seeds to isolate dataset variance and training-run variance.

Chapter 7

Conclusion and Future Work

This project investigated whether a multi-modal, temporally-aware machine learning gatekeeper could improve eavesdropper detection in BB84 quantum key distribution networks, improving on existing implementations shown in current literature, with a focus on an adaptive eavesdropper who suppresses their channel signatures in an attempt to evade detection.

Both hypotheses stated were supported by the primary fixed-dataset evaluation, with a statistically significant advantage over Logistic Regression and Random Forest ($p = 0.0007$ and $p = 0.0001$ respectively). However, the stratified evaluation motivated by our second hypothesis showed that the aggregate evaluation is not the only metric that should be considered when comparing models, as it conceals a substantial advantage for the LSTM when dealing with stealthy attacks. The LSTM only degraded by 9.0% when looking at the stealthy attack subtype, whereas Logistic Regression and Random Forest dropped by 16.3% and 35.8% respectively. No prior research evaluates detection of an attacker that suppresses their signature on the quantum channel; this research shows that a multi-modal model provides a measurable advantage in aggregate on a fixed-dataset, and a substantially larger and more consistently generalisable one precisely where security is most at risk.

When hypothesis testing was performed in the cross-dataset evaluation, the LSTM's aggregate ROC-AUC advantage over Random Forest did not consistently hold ($p = 0.9274$), although it kept its advantage over Logistic Regression ($p = 0.0107$). However, its stealthy-attack recall remained statistically significant against both baseline models, with $p = 0.0097$ and $p = 0.0108$ respectively. This shows that the second hypothesis generalises robustly to unseen datasets, rather than ones the model has already been trained on.

Discovering this advantage for the LSTM led to a secondary contribution: that aggregate accuracy and ROC-AUC alone are inadequate metrics for evaluating eavesdropper detectors when the channel is saturated with easily detected attacks. Sorting attacks into subtypes when evaluating gatekeeper models is proposed as a necessary practice for future QKD anomaly-detection research, as aggregate scores can mask differences in model robustness

that are most important for real-world application.

This project demonstrates that a multi-modal, temporally aware gatekeeper provides a statistically significant advantage in detecting adaptive, signature-suppressed eavesdropping in BB84 quantum key distribution, and that this advantage is amplified precisely against the strategy an adaptive eavesdropper is most likely to use.

Reflecting on the project as a whole, we have learned that some results can be misleading and hide underlying differences between models, meaning a more thorough evaluation of results can distinguish the advantages between models. Additionally, the test performed on a model must be generalised and robust to understand if a model's advantage is significant or if it is governed by a favourable test set. Throughout this process we had to develop an understanding of paired statistical significance testing, which proved to be essential for evaluating and comparing model performance, rather than relying on face-value estimates alone. We have also learned about the iterative nature of experimental design and understanding that more often than not, initial methodologies are often flawed, but they serve as stepping stone to continuously adapt and refine a final framework. If we were to do this project again, we would begin with the cross-dataset evaluation and build on this methodology to test for different adaptive attack strategies, as well as test on a wider array of models and baseline benchmarks.

7.1 Future Work

7.1.1 Varied Attack Type Scope

This project only tested against the intercept-resend attack type; however, there are many other types of attack that an ideal gatekeeper would need to be successful at identifying, such as photon-number-splitting (PNS), Trojan-horse and detector-blinding attacks [25, 16]. These attacks affect different hardware signatures and would require a model that can monitor additional telemetry features to detect reliably.

7.1.2 Adaptive Attack Strategy

The attack strategy used in this framework injected suppressed-signature attacks at a fixed and randomly assigned intensity per episode, but a stronger and more realistic threat model would use a dynamic suppression strategy in response to detector feedback, which could be implemented in future work [35].

7.1.3 Real Quantum Channel Telemetry Data

Synthetic telemetry data was used, guided by real behaviours and phenomena, but this could be improved upon through either the introduction of behaviours seen in practical channels,

such as loss due to channel distance, or through the collection of real telemetry data allowing tracking of the qubit as it is sent along the channel and how the quantum hardware measures the pulse. This would further test the robustness of the proposed framework for detection.

7.1.4 Systematic Hyperparameter Sensitivity Analysis

As noted in Chapter 4, the hyperparameters to train all three models were largely based on standard defaults or, in the case of the baseline models, based on literature, rather than through extensive parameter analysis and testing. Future work could carry out a systematic sensitivity analysis across the learning rate, batch size, hidden dimensions, and dataset size to test how they affect the performance gap between the LSTM and the classical baseline models.

7.1.5 Improved Cross-Dataset Evaluation

There are several ways in which the cross-dataset evaluation could be improved. Firstly, increasing the number of training and test datasets could yield interesting results and further reinforce our findings. In addition, as noted in Chapter 6, each model is trained once on the pooled training data, and then evaluated on eight separate unseen datasets; however, future work could investigate this protocol across multiple training seeds in the same way as the fixed-dataset evaluation, allowing training-run variance to be isolated from dataset variance.

Appendix A

Project Specification

This appendix reproduces the original project specification, as prepared and agreed with the supervisor at the start of the project period.

Project Details

Project name: AI-Enhanced Quantum Cryptography

Host Organisation: Lancaster University

Project description:

The project aims to explore the integration of AI with Quantum Key Distribution (QKD) to enhance secure communication systems. Quantum Key Distribution, which uses quantum mechanics to securely exchange cryptographic keys, faces challenges in scalability and robustness. By incorporating AI algorithms, such as machine learning models, the project seeks to optimize key distribution protocols, improve error correction, and enhance security in the face of quantum attacks. This research could pave the way for more efficient and secure quantum communications, crucial for the future of cybersecurity.

For background, see Lo, H.-K., Curty, M., & Qi, B. (2014). Secure quantum key distribution. <https://doi.org/10.1038/nphoton.2014.134>; <https://www.nature.com/articles/s41598-022-12647-x>

Student deliverables: Product prototypes

Student's line manager: Richard Jiang

Project location(s): On campus

Student Requirements

Essential technical skills: Python, Qiskit, Neural Networks

Desirable technical skills: As above

Essential general skills: Good academic writing

Desirable general skills: As above

Essential experience: Data Science Research

Desirable experience: As above

Organisation Details

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